Indian Institute of Information Technology Allahabad
Quiz-2 Question Paper
Course Name: Database Management Systems
Program Name(s): BTech(IT) & BTech(IT-Bin)
Session: Jan-July 2025
Course Instructors: Dr. Anjali Gautam, Dr. S. Maity
Exam Date: 23-April-2025

Duration: 1 Hr **Full Marks: 20**

1. Assume (for simplicity) that only one tuple fits in a block and memory holds at most 3 blocks. Show the runs created on each pass of the sort-merge algorithm, when applied to sort the following tuples on the first attribute: (kangaroo, 17), (wallaby, 21), (emu, 1), (wombat, 13), (platypus, 3), (lion, 8), (warthog, 4), (zebra, 11), (meerkat, 6), (hyena, 9), (hornbill, 2), (baboon, 12).

[2.5 Marks]

2. Write the query equivalent to $\sigma_{salary<75000}(\prod_{salary}(instructor))$.

[2.5 Marks]

3. Write the role of parser and translator in query processing.

[2.5 Marks]

4. Write the cost estimate of A1 linear search algorithm of selection operation, by considering only the number of block transfers from disk and the number of seeks as the cost measures for the simplicity.

[2.5 Marks]

5. A transaction performs the following operations:
T1: Read(A), A = A + 100, Write(A), Read(B), B = B - 100, Write(B)
Explain how **Atomicity** and **Durability** affect the execution of this transaction in the event of a system crash.

[2.5 Marks]

6. Explain the following **isolation anomalies** with suitable transaction examples:
a) Dirty Read b) Non-repeatable Read c) Phantom Read
Also, map them to isolation levels.

[2.5 Marks]

7. Consider a set of transactions scheduled using **Two-Phase Locking (2PL)**. Demonstrate how 2PL ensures serializability.

[2.5 Marks]

8. Draw the lock tree and explain how the **Tree Protocol** avoids deadlock, using an example of a file directory hierarchy.

[2.5 Marks]

**********END**********

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End Sem Question Paper

Course Name: Database Management System Course Instructor/ Coordinator: Dr. S. Maity and Dr. Anjali Gautam

Course Code: IT2009 (Core) Program Name(s): BTech (IT, BI) 4th Sem Exam Date: 12/05/2025 MM: 40

Note: Attempt all questions.

Q1. Suppose we have two tables: Customers and Orders. Find customers' names and their IDs who have placed at least one order using SQL query. Use the EXISTS operator. [2]

Q2. Write the PL/SQL code to update a table by increasing the salary of each employee by 1500. After the update, the implicit cursor attribute is used to find out how many rows were affected by the operation. [2]

Q3. Draw the ER Diagram for the hospital management system where Patient is attended by a Receptionist for registration and maintaining all his/her records, and allot them the time to meet a doctor. Doctor can be a trainee, visiting doctor or a permanent doctor. Once consulted by a doctor, a medical report is generated and based on that report if required the patient will be shifted to the hospital ward otherwise will be relieved by giving medicines after receiving the bill amount. All the hospital wards are governed by nurses to give timely medicines to patients.

A. The ER diagram should contain all the above mentioned details, with proper use of multivalued, composite, derived attributes, identify weak entity sets, and recursive relations. [3]

B. Write the relational algebra to display the list of patients registered in the hospital. [1]

C. Identify the highest paid doctor of the hospital using relational algebra. [1]

Q4. Let relations r1(A, B,C) and r2(C, D, E) have the following properties: r1 has 4,00 tuples, r2 has 750 tuples, 5 tuples of r1 fit on one block, and 6 tuples of r2 fit on one block. Estimate the number of block transfers and seeks required, using each of the following join strategies for r1⋈r2: (Size of memory = M) [4]

1. Nested-loop join 2. Block nested-loop join 3. Merge join 4. Hash join

Q5. Identify if any kind of anomaly exists in the mentioned schedule, if it is then what kind of anomaly it is?

T_1	T_2
lock-S(A)	
read(A)	
unlock(A)	
	lock-X(A)
	read(A)
	write(A)
	unlock(A)
	commit
lock-X(A)	
write(A)	
unlock(A)	
commit	

[1]

Q6. A schedule of 4 database transactions T1, T2, T3 and T4 is given below. Ri(A) and Wi(A) denote read and write of data item A by Transaction Ti, i=1,2,3,4. The transaction T1 aborts at the end.

S = R1(X) W1(Y) R2(X) R2(Y) R3(Z) R3(X) W3(Z) R4(Y) R4(X) R4(Z) ABORT(T1)

Which other transaction(s) will be required to be rolled back and why? [1.5]

Q7. What is view serializability and what are the desired conditions to make schedule S to be view equivalent to S' when they have the same set of transactions? [2.5]

Q8. Consider the following two statements about database transaction schedules:

A. Strict two-phase locking protocol generates conflict serializable schedules that are also recoverable.

B. Timestamp-ordering concurrency control protocol with Thomas Write Rule can generate view serializable schedules that are not conflict serializable.

Answer the above statements in terms of TRUE or False with proper justification. [2]

Q9. Define multivalued dependency. [1]

Q10. What is the Phantom Problem? How does Next-Key Locking differ from simple Index Locking in solving the Phantom Problem? Illustrate with an example. [1 + 1 = 2]

Q11. (a) A University Database implements Multiple Granularity Locking. A transaction needs to update the records of all students enrolled in a course. Explain the locking sequence it must follow.

(b) What problems can arise if intention locks are not properly used?

[1 + 2 = 3]

Q12. (a) Explain with an example how Deadlock Avoidance works using Wait-for Graph dynamic detection.

(b) What are the trade-offs between Deadlock Prevention and Deadlock Detection approaches in DBMS?

[1 + 1 = 2]

Q13. Consider the relations $r_1(A,B,C)$, $r_2(C,D,E)$, and $r_3(E,F)$, with primary keys A, C, and E, respectively. Assume that r_1 has 1000 tuples, r_2 has 1500 tuples, and r_3 has 750 tuples. Estimate the size of $r_1 \bowtie r_2 \bowtie r_3$. Justify your answer. [2]

Q14. Consider the relations $r_1(A,B,C)$, $r_2(C,D,E)$, and $r_3(E,F)$ of the above question. Assume that there are no primary keys, except the entire schema. Let $V(C, r_1)$ be 900, $V(C, r_2)$ be 1100, $V(E, r_2)$ be 50, and $V(E, r_3)$ be 100. Assume that r_1 has 1000 tuples, r_2 has 1500 tuples, and r_3 has 750 tuples. Estimate the size of $r_1 \bowtie r_2 \bowtie r_3$. Justify your answer. [2]

Q15. Suppose you are designing a query optimizer in which you have included only the following four equivalence rules into the optimizer:-

$$\sigma_{\theta 1 \wedge \theta 2}(E) = \sigma_{\theta 1}(\sigma_{\theta 2}(E))$$
$$\sigma_{\theta 1}(\sigma_{\theta 2}(E)) = \sigma_{\theta 2}(\sigma_{\theta 1}(E))$$
$$\sigma_{\theta}(E_1 \times E_2) = (E_1) \bowtie_{\theta} (E_2)$$
$$\sigma_{\theta 2}((E_1) \bowtie_{\theta 1} (E_2)) = (E_2) \bowtie_{\theta 1 \wedge \theta 2} (E_1)$$

Do you think the set is minimal? Explain your answer with proper justifications. [2]

Q16. Prove that any relation (table) with only two attributes is always in BCNF. [2]

Q17. Consider the following set of FDs defined over a relation schema R(A, B, C, D, E, F) :

{B→C, AB→D, CD→EF}

a) Derive all possible candidate keys of R.

b) Is the relation R under the given set of FDs in BCNF? Properly explain your answer. [1 + 1 = 2]

Q18. Consider the following set of functional dependencies on a relation R(A,B,C):-

F = {A→BC, B→C, AB→C}

Is there any extra FD or, is there any extraneous attribute in any of the FDs? Justify your answer. [2]

END

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Mid Semester Examination Question Paper

Course Name: Database Management System **Course Instructor/ Co-ordinator:** Section A = (ANG) Dr. Anjali Gautam, Section B = (SDM) Dr. Soumyadev Maity, Section C = (RNV) Dr. Ranjana Vyas

Course Code: PC-IT-DBM-404 **Program Name(s):** B.Tech (IT & IT-Bin) 4th sem.& 6th Sem (back) **Exam Date:** 28.02.2024 **MM:** 25

Schema Diagram for University Database

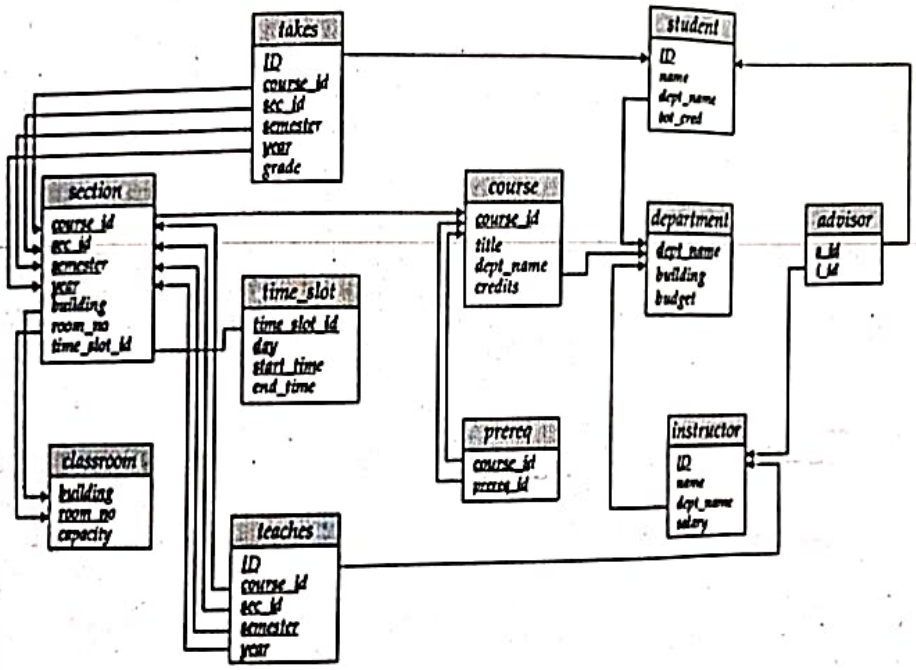

Figure-1

Q. 1. Using the university schema given in Figure-1, write the following query in:-

1. relational algebra
2. SQL

Query: "Find the course_ids and titles of all courses offered by the "IT" department that have at least 3 credits."

2x1 = 2 Marks

Q 2. Using the university schema given in Figure-1, write the following query in relational algebra:
"Find the names of all students who have taken at least one course offered by the "IT" department."

1 Marks

Q.3. Define and elaborate the natural join with its' role in RDBMS querying mechanism.

OR,

Explain how a schema may contain redundancy if it violates 1 NF.

1 Marks

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Q. 4. Considering the below **school** table, answer the following questions.

Teacher_ID	Subject	Teacher_Age
25	Chemistry. Biology	30
47	English	35
83	Math, Computer	38
65	German	32
77	DBMS, DS	40

1. Give the reason for the above table not in 1NF criteria, with proper explanation. **1.5 Marks**
2. Redesign the above 'school' table with the new table name 'updated_school' to align it with the conditions of 1NF of the relational database. **1.5 Marks**

Q.5. Consider the following employee database where the primary keys are underlined. Give an expression in SQL for each of the following queries. **2 Marks**

employee (<u>employee_name</u>, street, city)
works (<u>employee_name</u>, company name, salary)
company (<u>company_name</u>, city)
manages (<u>employee_name</u>, manager name)

a) Find the names, street addresses, and cities of residence of all employees who work for "First Bank Corporation" and earn more than $10,000.
b) Find all employees in the database who do not work for "First Bank Corporation".

Q. 6. Write relational algebra expression corresponding to the following SQL statement:- **0.5 Marks**

```
select A1, A2, sum(A3)
from r1, r2, ..., rm
where θ
group by A1, A2
```

Q. 7. Prove the pseudo-transitivity rule on FDs using Armstrong's Axioms. Clearly show which axiom you apply where. **0.5 Marks**

Q. 8. Explain one relative advantage of the BCNF normal form over 3NF and one advantage of 3NF over BCNF. **1 Marks**

Q. 9. Prove that any relation (table) with only two attributes is always in BCNF. **1 Marks**

Q. 10. "3NF decomposition algorithm (which you have studied) ensures that the decomposition is always i) lossless and ii) dependency preserving". Explain whether you agree with this statement or not. **1 Marks**

Q. 11. Consider the following set of FDs defined over a relation schema R(A, B, C, D, E, F) : **1x5=5 Marks**

$$\{B \rightarrow C,\ AB \rightarrow D,\ CD \rightarrow EF\}$$

a) Derive all possible candidate keys of R.
b) Is the reaction R under the given set of FDs in BCNF? Properly explain your answer.
c) If not in BCNF, perform a BCNF decomposition. Is your decomposition dependency preserving? Justify your answer.
d) Is the reaction R under the given set of FDs in 3NF? Properly explain your answer.

c) If not in 3NF, perform a 3NF decomposition.

Q. 12. Consider the following set of FDs defined over a relation schema R(A, B, C, D, E, F) :

$$\{A \rightarrow BCD,\ BC \rightarrow DE,\ B \rightarrow D,\ D \rightarrow A\}$$

Compute a canonical cover for the above set of FDs. **1 Marks**

Q 13. Assume we have the following application that models soccer teams, the games they play, and the players in each team. In the design, we want to capture the following:

- We have a set of teams, each team has an ID (unique identifier), name, main stadium, and to which city this team belongs.
- Each team has many players, and each player belongs to one team. Each player has a number (unique identifier), name, DoB, start year, and shirt number that he uses.
- Teams play matches, in each match there is a host team and a guest team. The match t
  the stadium of the host team.
- For each match we need to keep track of the following:
  - The date on which the game is played
  - The final result of the match
  - The players participated in the match. For each player, how many goals he scored, whether or not he took yellow card, and whether or not he took red card.
  - During the match, one player may substitute another player. We want to capture this substitution and the time at which it took place.
- Each match has exactly three referees. For each referee we have an ID (unique identifier), name, DoB, years of experience. One referee is the main referee and the other two are assistant referees.

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Design an ER diagram to capture the above requirements. State any assumptions you have that affect your design (use the back of the page if needed). Make sure cardinalities and primary keys are clear.

3 Marks

Note: A yellow card signifies an official warning for the individual while a red card means the individual must leave the area of play.

Q 14. Map the ER Diagram in Question 13 to create the relational model corresponding to the described application. Basically, list the CREATE TABLE statements with the attribute names, and appropriate data types. Also make sure to have the primary keys and foreign keys clearly defined. **2 Marks**

Q 15. Consider the following relations for a database that keeps track of business trips of salespersons in a sales office:

SALESPERSON (SSN, Name, Start_Year, Dept_No)
TRIP (SSN, From_City, To_City, Departure_Date, Return_Date, Trip_ID)
EXPENSE (Trip_ID, Account#, Amount)

Specify the foreign keys for this schema, stating any assumptions you make. **1 Marks**

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End-Sem Question Paper

Course Name: *Database Management Systems* **Course Instructor/ Coordinator:** Dr. S. MAITY

Course Code: *PC-IT-DBM404* **Program Name(s):** *BTech(IT) & BTech(IT-Bin)*

Exam Date: *07-May-2024* **MM:** *40*

Answer All Questions.

Q1. Consider the following schedule for transactions T1, T2 and T3: **2 Marks**

T1	T2	T3
1 Read (X)		
	2 Read (Y)	
		3 Read (Y)
	4 Write (Y)	
5 Write (X)		
		6 Write (X)
	7 Read (X)	
	8 Write (X)	

Find out the correct serialization order of the transactions in the above schedule. Justify your answer, why so.

Q2. Consider the transactions T1, T2, and T3 and the schedules S1 and S2 given below.

T1: r1(X); r1(Z); w1(X); w1(Z)
T2: r2(Y); r2(Z); w2(Z)
T3: r3(Y); r3(X); w3(Y)
S1: r1(X); r3(Y); r3(X); r2(Y); r2(Z);
w3(Y); w2(Z); r1(Z); w1(X); w1(Z)
S2: r1(X); r3(Y); r2(Y); r3(X); r1(Z);
r2(Z); w3(Y); w1(X); w2(Z); w1(Z)

1) Test the conflict serializability of the above schedules. **2 Marks**
2) Draw the precedence graph for both schedules. **1 Marks**
3) The schedule which is conflict serializable, write the serializability order of the transactions for that schedule. **2 Marks**

Q3. (a) Explain about two phase locking (2PL) protocol.
(b) For the schedule given below

T1	T2
Read A	
	Read B
Write A	

Read A
Write A
Write B

Read B
Write B

1. Find whether the above schedule is serializable or not? Justify your answer. **1 Marks**
2. Also find whether it can occur in the 2PL locking protocol? Justify your answer. **2 Marks**

Q4. Explain view serializability. In the given schedule, check whether it is view serializable or not. Also check whether the schedule is conflict equivalent to <T1, T5>. In both cases, justify your answer. **2 Marks**

T_1	T_5
read(A)	
$A := A - 50$	
write(A)	
	read(B)
	$B := B - 10$
	write(B)
read(B)	
$B := B + 50$	
write(B)	
	read(A)
	$A := A + 10$
	write(A)

Q5. During its execution, a transaction passes through several states, until it finally commits or aborts. List all possible sequences of states through which a transaction may pass. Explain why each state transition may occur. **2 Marks**

Q6. Explain the distinction between the terms serial schedule and serializable schedule. **1 Marks**

Q.7 Most implementations of database systems use strict two-phase locking. Suggest three reasons for the popularity of this protocol. **3 Marks**

Q.8 Underwhat conditions is it less expensive to avoid deadlock than to allow deadlocks to occur and then to detect them? non prem **3 Marks**

Q.9. If deadlock is avoided by deadlock-avoidance schemes, Is starvation still possible? Explain your answer. **2 Marks**

Q.10 In multiple-granularity locking, what is the difference between implicit and explicit locking **3 Marks**

Q11.

A) Let r and s be the two instances of the schema R(A, B, C) and S(A, D, E), where attribute A in S is a foreign-key referencing to schema R. For the relational algebra expression E, shown below, estimate the size (in terms of no. of tuples) of the result:-

$$E: \sigma_{\theta}(r \bowtie (\Pi_{L}(s)))$$
$$L = \{A, D\}$$
$$\theta = \theta_1 \vee \theta_2$$
$$\theta_1: D > 70$$
$$\theta_2: A = a_1$$

The database catalog info is given below:-

Total no. of tuples in $r = n_r$

Total no. of tuples in $s = n_s$

Distinct values of attribute A in $r = \{a_1, a_2, a_3, a_4\}$

$Max(D, s) = 100$

$Min(D, s) = 10$

Note that, $size(E_1) \neq size(E_2)$ even if $E_1 \equiv E_2$.

4 Marks

B) Estimate the size of the above expression E in bytes if:-

n_r=100, n_s=80,

size-of(A) = 1 byte,

size-of(B) = 2 bytes, size-of(C) = 2 bytes, size-of(D) = 2 bytes, size-of(E) = 2 bytes

0.5 Marks

C) Draw the expression tree corresponding to the above mentioned relational algebra expression E.

0.5 Marks

D) What would be the optimal equivalent expression corresponding to the above expression E if we use a heuristic rules based query optimizer? Draw the expression tree corresponding to your answer.

1 Marks

Q12. "Perform selection early" and "perform projection early" are the two most commonly used heuristic rules used by many heuristic based query optimizers. Give an example to clearly demonstrate that performing projection early may not always give you an optimized query evaluation plan.

2 Marks

Q13. In order to reduce the time complexity of the query optimization process, many query optimizers take the advantage of dynamic programming. Explain how a query optimizer can use dynamic programming to take this advantage, with an example.

2 Marks

Q14. What is a left-deep join tree? In which scenario a query optimizer would prefer to choose left-deep or right-deep trees over other equivalent expressions? What is the advantage of using left-deep or right-deep trees over other equivalent expressions?

2 Marks

Q15. If you think that any of the following equivalence rules is incorrect then give a counterexample (instances our relations) to show that the LHS and the RHS of the expression or not equivalent:-

(i) $\sigma_\theta(E_1 - E_2) = \sigma_\theta(E_1) - \sigma_\theta(E_2)$

(ii) $\Pi_L(E_1 \cup E_2) = \Pi_L(E_1) \cup \Pi_L(E_2)$

(iii) $\Pi_L(E_1 - E_2) = \Pi_L(E_1) - \Pi_L(E_2)$

(iv) $\sigma_\theta(E_1 - E_2) = \sigma_\theta(E_1) - E_2$

2 Marks

END

C2: Quiz-1: Database Management Systems (DBMS)

Session: Jan-July 2023, B.Tech(I.T.), Section-B
Date: 31-March-2023; Duration of Exam: 1 Hour
Course Instructor: Dr. S. Maity **Full Marks: 60**

IMPORTANT INSTRUCTION: You MUST show all steps of your calculations clearly. Otherwise, you will not get any marks even if your final answer is correct.

Q. 1. Let, R(M,N,O,P,Q,R,S) and F={ P → QS, OP → Q, OP → R, N → MO, M → O, O → S, MN → M }.

a) Is R in BCNF w.r.t F? If not, perform a BCNF decomposition. **NOTE:** Always target the first FD which is violating BCNF condition, for decomposition purpose.

b) Is R in 3NF w.r.t. F? If not, perform a 3NF decomposition.

10 x 2 = 20 marks

Q. 2. Prove that Armstrong's Axioms are sound.

10 marks

Q. 3. State the definition of multivalued dependency.

5 marks

Q. 4. State the definition of 4NF.

5 marks

Q. 5. Consider the following Project-Advisor schema:-

Advisor(Project_id, instructor_id, student_id, instructor_dept, student_dept)

Where the user imposed the following constraints:-

- Project_id is unique for each project.
- A student can take any number of guides from different departments but cannot take more than one guide from any one department.
- A student must take at-least one guide from his/her own department.
- A faculty can work under only one department but can guide students of other departments.
- A faculty can guide maximum 5 students irrespective of departments.

(a) Define the set of FDs imposed by the user (for ease of writing, you may assign a letter for every attribute, for instance: A, B, C, D, E for the five attributes of the schema, and denote the schema as R).

(b) Is the relation in BCNF or 3 NF? Justify your answer.

(c) Now, if you remove the first attribute (Project_id) from the above schema then do you think that the modified schema would be in BCNF, 3 NF? Justify your answer.

(d) If you think the answer to the previous question (5.c) is no, then do a 3 NF decomposition of the modified schema.

10 x 2 = 20 marks

END

C2: Quiz-1: Database Management Systems (DBMS)

Session: Jan-July 2023, B.Tech(I.T.), Section-B
Date: 31-March-2023; Duration of Exam: 1 Hour
Course Instructor: Dr. S. Maity **Full Marks: 60**

IMPORTANT INSTRUCTION: You MUST show all steps of your calculations clearly, Otherwise, you will not get any marks even if your final answer is correct.

Q. 1. Let, R(M,N,O,P,Q,R,S) and F={ P → QS, OP → Q, OP → R, N → MO, M → O, O → S, MN → M }.

a) Is R in BCNF w.r.t F? If not, perform a BCNF decomposition. **NOTE:** Always target the first FD which is violating BCNF condition, for decomposition purpose.
b) Is R in 3NF w.r.t. F? If not, perform a 3NF decomposition.

10 x 2 = 20 marks

Q. 2. Prove that Armstrong's Axioms are sound.

10 marks

Q. 3. State the definition of multivalued dependency.

5 marks

Q. 4. State the definition of 4NF.

5 marks

Q. 5. Consider the following Project-Advisor schema:-

Advisor(Project_id, instructor_id, student_id, instructor_dept, student_dept)

Where the user imposed the following constraints:-

- Project_id is unique for each project.
- A student can take any number of guides from different departments but cannot take more than one guide from any one department.
- A student must take at-least one guide from his/her own department.
- A faculty can work under only one department but can guide students of other departments.
- A faculty can guide maximum 5 students irrespective of departments.

(a) Define the set of FDs imposed by the user (for ease of writing, you may assign a letter for every attribute, for instance: A, B, C D, E for the five attributes of the schema, and denote the schema as R).
(b) Is the relation in BCNF or 3 NF? Justify your answer.
(c) Now, if you remove the first attribute (Project_id) from the above schema then do you think that the modified schema would be in BCNF, 3 NF? Justify your answer.
(d) If you think the answer to the previous question (5.c) is no, then do a 3 NF decomposition of the modified schema.

10 x 2 = 20 marks

END

INDIAN INSTITUTE OF TECHNOLOGY ALLAHABAD

DATABASE MANAGEMENT SYSTEM

C3 REVIEW TEST

Date: 19/05/2023 **Time: 2 Hours** **Max. Marks: 50**

Q1. Consider a database used to record the marks that students get in different exams of different course offerings (sections).

a. Construct an E-R diagram that models exams as entities, and uses a ternary relationship, for the database.

b. Construct an alternative E-R diagram that uses only a binary relationship between *student* and *section*. Make sure that only one relationship exists between a particular *student* and *section* pair, yet you can represent the marks that a student gets in different exams.

(Hint: three main entities are student, section and course, others may be or not depending on your perception from the questions.)

5 marks

Q2. Write the following inserts, deletes or updates in SQL, using the university schema. **5 marks**

a. Increase the salary of each instructor in the Comp. Sci. department by 10%.

b. Delete all courses that have never been offered (that is, do not occur in the *section* relation).

c. Insert every student whose *tot_cred* attribute is greater than 100 as an instructor in the same department, with a salary of Rs. 10,000.

(*SQL data definitions:*

```
create table department
    (dept_name    varchar (20)
    building      varchar (15)
    budget        numeric (12,2)
    primary key (dept_name))

create table course
    (course_id    varchar (7),
    title         varchar (50),
    dept_name     varchar (20),
    credits       numeric (2,0),
    primary key (course_id),
    foreign key (dept_name) references department),

create table student
    (ID           varchar (5),
    name          varchar (20) not null,
    dept_name     varchar (20),
    tot_cred      numeric (3,0) default 0,
    primary key (ID));

create table instructor
    (ID           varchar (5),
    name          varchar (20) not null,
    dept_name     varchar (20),
    salary        numeric (8,2),
    primary key (ID),
    foreign key (dept_name) references department),

create table section
    (course_id    varchar (8)
    sec_id        varchar (8),
    semester      varchar (6),
    year          numeric (4,0),
    building      varchar (15),
    room_number   varchar (7),
    time_slot_id  varchar (4),
    primary key (course_id, sec_id, semester, year),
    foreign key (course_id) references course).
```

)

Q3. Suppose that we have a relation *marks (ID, score)* and we wish to assign grades to students based on the score as follows: grade F if *score* < 40, grade C if 40 ≤ *score* < 60, grade B if 60 ≤ *score* < 80, and grade A if 80 ≤ *score*. Write SQL queries to do the following: **5 marks**

a. Display the grade for each student, based on the marks relation.

b. Find the number of students with each grade.

Q4. Suppose that we decompose the schema R = (A, B, C, D, E) into r1(A, B, C) and r2(A, D, E). Show that this decomposition is a lossless decomposition if the following set F of functional dependencies holds:

$A \rightarrow BC \qquad CD \rightarrow E \qquad B \rightarrow D \qquad E \rightarrow A$ **5 marks**

Q5. A functional dependency $\alpha \rightarrow \beta$ is called a partial dependency if there is a proper subset γ of α such that $\gamma \rightarrow \beta$. We say that β is partially dependent on α. A relation schema R is in second normal form (2NF) if each attribute A in R meets one of the following criteria:

- It appears in a candidate key.
- It is not partially dependent on a candidate key.

Show that every 3NF schema is in 2NF. **5 marks**

Q6. Given a relation R(A,B,C,D) and Functional Dependency set FD = { $AB \rightarrow CD$, $B \rightarrow C$ }, determine whether the given R is in 2NF? If not, convert it into BCNF. **3 marks**

Q7. Consider the following *marks-sheet* table which keeps the marks of all students of the DBMS course for the current semester:- **4 marks**

marks-sheet (enroll-no varchar2(20), sec-id varchar2(20), marks number)

Assume the following:-

- there are total 100 students registered for the DBMS course, who are equally divided into 5 sections
- the minimum and maximum marks obtained by the students over all sections is 40 and 100 respectively
- the enroll-no.s of the students are of the form "IIT2021xxx"

- one character takes 1 byte of storage
- marks are uniformly distributed between the min. and max. marks

Estimate the size of the result of the following queries,:-

i) "Find the enroll-no.s of all students who belong to Sec-B and obtained less than 70 marks".

ii) "Find the enroll-no.s of all students who belong to Sec-B or, who have obtained more than 70 marks".

Q8. If you think that any of the following equivalence rules is incorrect then give a counterexample (instances our relations) to show that the LHS and the RHS of the expression or not equivalent:- **4 marks**

$$(i)\ \sigma_\theta(E_1 - E_2) = \sigma_\theta(E_1) - \sigma_\theta(E_2)$$

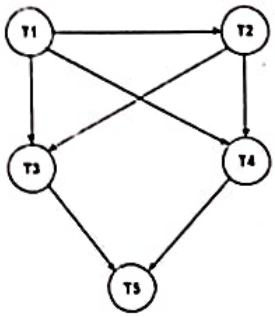
$$(ii)\ \Pi_L(E_1 \cup E_2) = \Pi_L(E_1) \cup \Pi_L(E_2)$$

$$(iii)\ \Pi_L(E_1 - E_2) = \Pi_L(E_1) - \Pi_L(E_2)$$

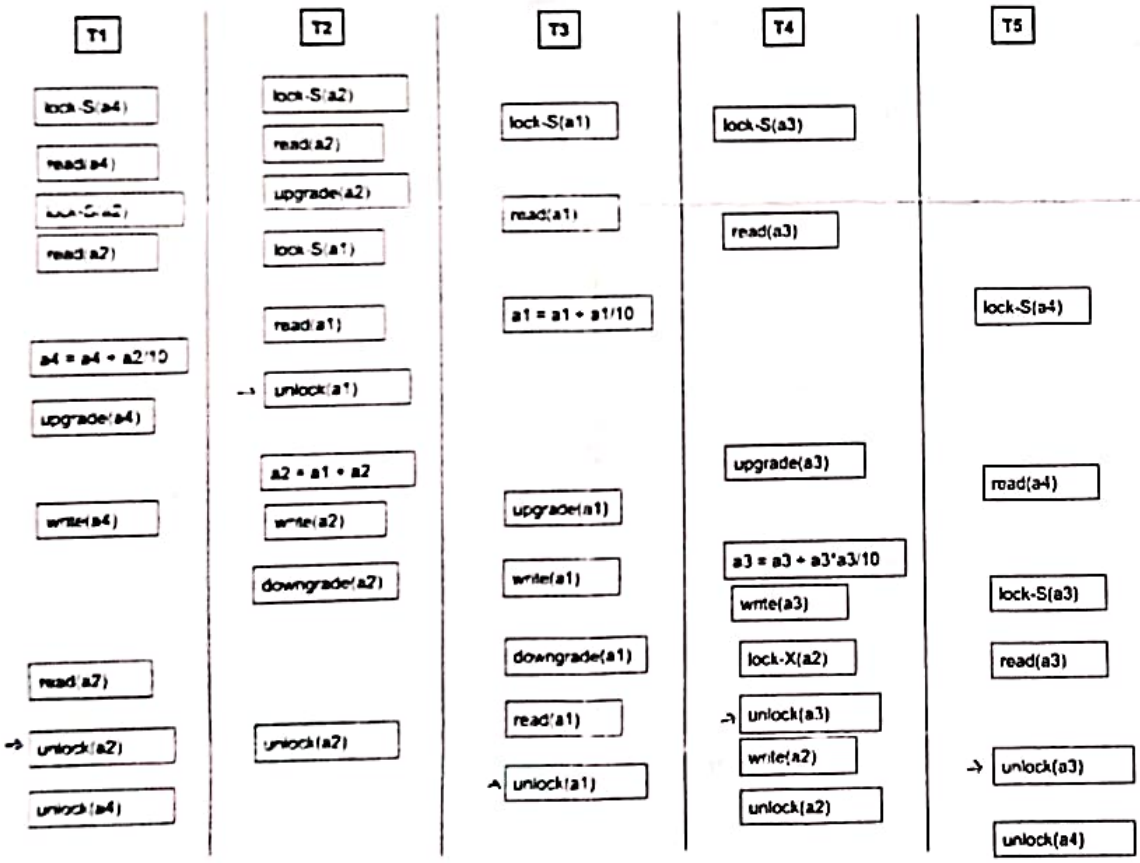
$$(iv)\ \sigma_\theta(E_1 - E_2) = \sigma_\theta(E_1) - E_2$$

Q9. Consider the following precedence graph, corresponding to a schedule S:-
Do you think that schedule S is serializable? If so, then what would be the possible conflict equivalent serial schedules corresponding to S?

3 marks



Q10. Do you think that the following schedule is legal under 2PL protocol? If yes, then what is the conflict equivalent serial schedule of the given schedule? **5 marks**



Q11. Answer the following questions:-

1. State the timestamp ordering protocol (used for consistency control). If we incorporate the Thomas' write rule in timestamp ordering protocol then which of the following is / are true?
   - The schedules allowed by the protocol will always be conflict serializable.
   - The schedules allowed by the protocol will always be view serializable.
   - A conflict serializable schedule will always be allowed by the protocol.
   - A view serializable schedule will always be allowed by the protocol.
2. Query optimizer generates optimized evaluation plan for a given query expression. What is an "evaluation plan"?
3. Explain the difference between the terms *serial schedule* and *serializable schedule*.

2 marks x 3 = 6 marks

Database Management Systems (DBMS) - QUIZ

Session JAN-JULY 2024

Section-B; Course Instructor: Dr. S. Maity

Full Marks: 25 **Duration: 1 Hour**

Q. 1. Let the following relation schemas be given:

R = (A, B, C)

S = (D, E, F)

Let relations r(R) and s(S) be given. Give an expression in the tuple relational calculus that is equivalent to the following algebra expression:-

$$\Pi_{A,F}(\sigma_{C=D}(r \times s))$$

2 Marks

Q. 2. Let R = (A, B) and S = (A, C), and let r (R) and s(S) be relations. Write expressions in relational algebra for the following query:-

$$\{< a > \mid \exists b(< a, b > \in r \ \wedge \ b = 7)\}$$

2 Marks

Q. 3. State the ACID properties. The concurrency control protocols are responsible for ensuring which of the ACID properties?

2 + 0.5 = 2.5 Marks

Q. 4. Suppose we have 5 transactions T1, .., T5 each consisting of 10 instructions. How many serial schedules are possible with these transactions? How many parallel schedules are possible? What is the difference between a serial schedule and a serializable schedule?

0.5 + 0.5 + 1 = 2 Marks

Q. 5. For a fixed set of transactions (T1, ..., Tn) all serial schedules are conflict equivalent to each other. Prove whether the statement is correct or not.

2 Marks

Q. 6. State the essential condition that must be satisfied for a schedule to be recoverable. State the essential condition that must be satisfied for a schedule to be cascadeless.

1 + 1 = 2 Marks

Q. 7. Consider two transactions T1 and T2. Transaction T1 transfers Rs 50 from account A to account B. Transaction T2 transfers Rs 10 from account A to account B. Write a schedule with T1 and T2 which is serializable but neither conflict serializable nor view serializable. Explain your answer properly.

4 Marks

Q. 8. Consider the precedence graph shown in Figure 1. Is the corresponding schedule conflict serializable? If yes, then write all possible serial schedules which are conflict equivalent to the given schedule.

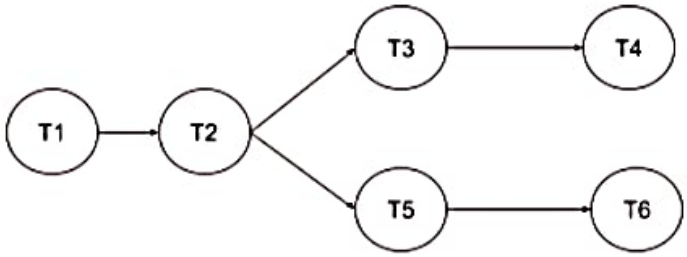


Figure-1

2 Marks

Q. 9. What are the different weaker levels of consistency? Why should a database system implement a weaker level of consistency?

2 + 0.5 = 2.5 Marks

Q. 10. State all the differences between the tree based locking protocol and the multiple granularity locking protocol.

4 Marks

— END —

Q1. Consider the following two transactions:

T_{34}: read(A);
read(B);
if $A = 0$ **then** $B := B + 1$;
write(B).

T_{35}: read(B);
read(A);
if $B = 0$ **then** $A := A + 1$;
write(A).

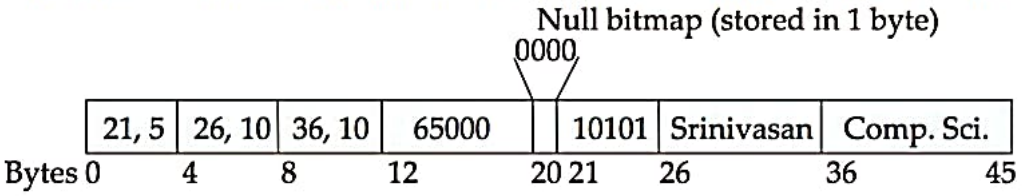
Add lock and unlock instructions to transactions T31 and T32 so that they observe the two-phase locking protocol. Can the execution of these transactions result in a deadlock?

Q2. What are different intention mode locks?

Q3. Consider the relations r 1(A, B, C), r 2(C, D, E), and r 3(E, F), with primary keys A, C, and E, respectively. Assume that r 1 has 1000 tuples, r 2 has 1500 tuples, and r 3 has 750 tuples. Estimate the size of $r1 \bowtie r2 \bowtie r3$.

DBMS C2 Quiz-2
Session: January – July 2020
Dr. S. Maity
Full Marks: 32

Q.1. Explain what is the meaning of {21, 5} and {26, 10} in the following format?



2 Marks

Q.2. How fixed-length file insertions and deletions are relatively simpler as compared to those for variable-length?

2 Marks

Q.3. How to organize fields within a record of fixed length?

2 Marks

Q.4. How can you minimize the number of disk-block transfers in a database system?

2 Marks

Q.5. When you need to replace a block in the buffer, which block would you choose to replace?

2 Marks

Q.6. What different strategies are available to organize records and the relations into files?

2 Marks

Q.7. What are the different ways in which variable-length records may arise in a database system?

4 Marks

Q.8. Suppose the following set F of functional dependencies are applicable on the relation schema *r*(S, T, U, V, W, X).

$$F=\{T \rightarrow V,\ TX \rightarrow VW,\ V \rightarrow U,\ U \rightarrow TVX\}$$

Answer the following questions showing clearly the steps for your calculations: -

i) Find the candidate keys for *r*.

ii) Compute a canonical cover F_c for the above set of functional dependencies F.

iii) With respect to your computed F_c, is the relation *r* in BCNF? Justify answer. If not, then make a BCNF decomposition.

[redacted]

iv) With respect to your computed F_c, is the relation *r* in 3NF? Justify answer. If not, then make a 3NF decomposition.

v) Explain whether your BCNF and 3NF decompositions are lossless or not.

3 + 4 + 3 + 3 + 3 = 16 Marks